Fiscal Year:	FY 2016	**Task Last Updated:**	FY 01/19/2016
PI Name:	Su, Ching-Hua Ph.D.		
Project Title:	Crystal Growth of II-VI Compound Semiconductors		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
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PI Email:	ching.h.su@nasa.gov	**Fax:**	FY 256-544-8762
PI Organization Type:	NASA CENTER	**Phone:**	256-544-7776
Organization Name:	NASA Marshall Space Flight Center		
PI Address 1:	Mail Code EM31		
PI Address 2:	Materials and Processing Laboratory		
PI Web Page:			
City:	Huntsville	**State:**	AL
Zip Code:	35812	**Congressional District:**	5
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No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Rogers, Jan	**Contact Phone:**	256.544.1081
Contact Email:	jan.r.rogers@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: Change in end date per P. Downey/MSFC (Ed., 9/28/2011)		
Key Personnel Changes/Previous PI:	Feb. 2013 report: Co-Investigator: Dr. Narayanan Ramachandran, Jacobs ESSSA Group, ER42, NASA Marshall Space Flight Center.		
COI Name (Institution):	Ramachandran, Narayanan (Jacobs ESSSA Group)		
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	Two reasons have been put forward to account for the improved crystal quality when crystals are grown in a microgravity environment. The first is weight-related reductions in crystal strain and defects. These are thought to be caused by the weight of the crystals during processing at elevated temperatures and retained on cooling, particularly for materials with low yield strength. The second and more general reason is related to the reduction in density-gradient driven convection in the fluid during crystal growth in reduced gravity. The objectives and experiment description for the proposed flight experiment “Growth of Ternary Compound Semiconductors” by (1) vapor growth and (2) melt growth are described below. 1a) Vapor Growth		

Task Description:

The physical vapor transport (PVT) crystal growth process consists of essentially three processes: sublimation of source material, transport of vapor species, and condensation of vapor species to form crystal. The latter two processes can be affected by the convection caused by gravitational accelerations on Earth. Reductions in such convection in low gravity is expected to yield a nearly diffusion-limited growth condition which results in more uniform growth rates (on the microscopic scale) and hence greater crystalline perfection and compositional homogeneity. This reduction of convective contamination in a reduced gravity environment will help to simplify the complexity of the coupled mass transport and growth kinetics problem. An improved comparison between the experimental results and theoretical simulations is also expected and will be beneficial to any growth process involving vapor mass transport or vapor-crystal interface growth kinetics.

The materials to be studied in this project are ZnSe and the related ternary compound semiconductors such as ZnSeTe, ZnSeS, and ZnCdSe. The scientific objectives of the growth of ZnSe crystals in reduced gravity using physical vapor transport (PVT) processes are:

• to establish the relative contributions of gravity-driven fluid flows to (i) the non-uniform incorporation of impurities and defects and (ii) the deviation from stoichiometry observed in the grown crystals as a result of buoyancy-driven convection and irregular fluid-flows caused by growth interface fluctuations and interface evolution.

• to assess self-induced strain effects developed during processing at elevated temperatures and retained on cooling caused by the weight of the crystals from x-ray topography and by measuring the density and distribution of various crystalline defects.

• to obtain high quality space-grown materials for various thermophysical and electrical properties measurements and as substrates for device fabrication and thus assess device performance as influenced by a substantial reduction in gravity-related effects.

1b) Melt Growth

The melt growth of homogeneous ternary compound semiconductors, such as cadmium zinc telluride (Cd1-xZnxTe, $0 < x < 1$) is difficult because the thermal and compositional gradients in the melt just ahead of the growth interface during directional solidification can cause substantial gravity-driven fluid flows that alter the relative distribution of the two (CdTe and ZnTe) binary components and introduce structural defects into the grown crystals. The scientific objectives of the growth of CdZnTe crystals in reduced gravity using directional solidification process are:

• to investigate quantitatively the influences of gravitationally dependent phenomena on the growth process and the incorporation of impurities and defects in the resulting crystals.

• to better understand and control the growth parameters in the production process on Earth so as to increase the yield of usable material from the grown ingot.

• to assess strain effects developed during processing at elevated temperatures and retained on cooling, caused by the weight of the crystals and, possibly, the contact with the growth container, from the structural characterization and by measuring the density and distribution of various crystalline defects.

• to obtain high quality space-grown materials for various electrical and optical property measurements and as materials for the x-ray and gamma-ray detector fabrication as well as substrates for infrared device application and thus assess device performance as influenced by a substantial reduction in gravity-related effects.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

• In crystal growth there is a need to understand the relation between processes in the fluid phase (both liquid and vapor), such as buoyancy driven convection and the incorporation of impurities, and defects in the resulting crystal
• Relation between fluid phase processes and the generation of defects in a grown crystal is an outstanding problem in materials growth

• Crystal quality greatly influences important electronic properties in materials • Ternary compound semiconductors are of vital national interest as sensors in x-ray telescopes and for homeland security, and as substrate materials for infrared sensors

• Studies in microgravity will be compared with modeling and will be used to optimize ground-based experiments

During this period, we have the following progresses on the categories of melt growth and the vapor growth of ternary II-VI compound semiconductors:
1. Melt growth:

• A New Technology Report (NTR), “A Method to Process High Efficiency Functional Gradient Thermoelectric Materials,” was submitted by Ching-Hua Su (EM31). Thermoelectric materials convert thermal energy, i.e., heat, into electric energy. The converting efficiency is usually rated by the values of Figure of Merit, zT, which varies as a function of temperature and shows a maximum at a certain temperature TM. Common thermoelectric materials have uniform composition and, when positioned in a thermal gradient, only a small portion of the material is operating in a narrow temperature range with high zT values. It is claimed, in this invention, that by carefully processing some compound semiconductors from the melts the resultant materials can possess thermoelectric properties such that TM within each small material layer varies monotonically along the axial direction of the processed ingot. This unique property of the so-called Functional Gradient Thermoelectric Material will improve the total converting efficiency of the thermoelectric component because, as the ingot is operating in a thermal gradient, each layer of the material along its axis is performing close to the temperature of the highest efficiency.

• A paper, “A Method of Promoting Single Crystal Yield during Melt Growth of Semiconductors by Directional Solidification,” by Ching-Hua Su (EM31) was published on Journal of Crystal Growth. Using the technology described in the patent US 8,535,440 “Method of Promoting Single Crystal growth during Melt Growth of Semiconductors” by Ching-Hua Su, the paper shows that the single crystal yield of large CdZnTe boules has been vastly improved from less than 30% to higher than 60%.

Task Progress:	• An article, “Mechanically Induced Nucleation Improves Crystalline Quality during Melt Growth of Semiconductors”, was published on NASA Tech Briefs, p. 44, Vol.39, No.2, February 2015. The article describes the recently awarded patent with a manufacturing technology that “significantly lower supercoiling results in the ideal growth condition of single crystal nucleation.” • A paper, “Thermal conductivity, electrical conductivity, and thermoelectric properties of CdTe and Cd0.8Zn0.2Te crystals between room temperature and 780oC,” by Ching-Hua Su/EM31, was published on AIP Advances. The thermal conductivity, electrical conductivity, and Seebeck coefficient of a vapor-grown CdTe and two melt-grown Cd0.8Zn0.2Te crystals, including one In-doped and one un-doped samples, were measured from room temperature to 780oC. The measured thermal conductivity provides the detailed information on the thermal fields of the growth ampoule/material assembly, which is critical for the design of growth process to achieve optimum thermal conditions for the growth of high crystalline quality crystals. Additionally, for the first time, the potential thermoelectric application of CdZnTe materials was evaluated from the three sets of measurements. • A presentation, “Characterization of CdZnTe single crystals grown under different cadmium overpressures,” was given by Ouloide Yannick Goue, Balaji Raghothomachar, Michael Dudley, (all at Stony Brook University) and Ching-Hua Su (EM31) at ACCGE-20, Big Sky, Montana. The single crystal of CdZnTe grown in-house at Marshall Space Flight Center (MSFC) has been characterized to assess its quality for the application as high-energy radiation detector. The micro defect of tellurium inclusion/precipitate can be detrimental to the detector performance as it reduces the carrier lifetime. Therefore, the crystal was grown by directional solidification under different controlled cadmium overpressure to adjust its stoichiometry. Using the results from synchrotron X-ray topography (SXRT) and infrared (IR) microscopy, the presentation focused on the correlation established between the formation of tellurium inclusion/precipitate and the cadmium overpressure applied during crystal growth. From this correlation, an optimum cadmium overpressure, which results in the least amount of tellurium inclusion/precipitate, has been determined. 2. Vapor growth: • An invited talk, “Crystal Growth of Ternary Compound Semiconductors in Low Gravity Environment,” was given by Ching-Hus Su (EM31) during the Collaborative Conference on Crystal Growth (3CG-2014) on November 6, 2014 in Phuket, Thailand. The presentation was focused on the preparations and results of ground-based crystal growth and characterization for the upcoming flight experiments on the International Space Station (ISS). • A New Technology Review (NTR), “Method to Improve Synthesis Process of High-Purity Bulk Multi-elements Compounds,” has been submitted by Ching-Hua Su (EM31). Multi-elements compounds have been used ubiquitously in various applications, ranging from the fields of electronics, optics, opto-electronics, thermoelectrics, superconductivity, to the recently developed application of spintronics. In many occasions of the typical process of synthesizing high-purity bulk multi-elements compounds, the resultant sample adheres to the fused quartz/silica ampoule during cooling and causes the interior cracks and, consequently, the inferior quality and low yield of the sample. This problem is especially prevalent when the more reactive elements, i.e., group I to IV in periodic table, are involved. It is claimed, in this invention, that by specifically preparing the fused quartz/silica ampoule and carefully processing the heating procedure the chemical reaction between the sample and the ampoule wall can be minimized and, at the same time, the reactions among elements maximized, so the crystalline quality of the resultant materials can be vastly improved.
Bibliography Type:	Description: (Last Updated: 02/18/2025)
Abstracts for Journals and Proceedings	Goue OY, Raghothomachar B, Dudley M, Su C-H. "Characterization of CdZnTe single crystals grown under different cadmium overpressures." Presented in the 20th American Conference on Crystal Growth and Epitaxy (ACCGE-20) and 17th U.S. Biennial Workshop on Organometallic Vapor Phase Epitaxy (OMVPE-17), Big Sky, MT, August 2-7, 2015. Abstracts of 20th American Conference on Crystal Growth and Epitaxy (ACCGE-20) and 17th U.S. Biennial Workshop on Organometallic Vapor Phase Epitaxy (OMVPE-17), Big Sky, MT, August 2-7, 2015. , Aug-2015
Abstracts for Journals and Proceedings	Su C-H. "Crystal Growth of Ternary Compound Semiconductors in Low Gravity Environment." Invited talk presented in the Collaborative Conference on Crystal Growth (3CG 2014), Phuket, Thailand, November 4-7, 2014. Abstracts of Collaborative Conference on Crystal Growth (3CG 2014), Phuket, Thailand, November 4-7, 2014. , Nov-2014
Articles in Other Journals or Periodicals	Article describing Ching-Hua Su's patent awarded in 2013 with a manufacturing technology that 'significantly lowers supercoiling results in the ideal growth condition of single crystal nucleation.' "Mechanically Induced Nucleation Improves Crystalline Quality during Melt Growth of Semiconductors." NASA Tech Briefs. 2015 Feb 1;39(2). http://www.techbriefs.com/component/content/article/7-ntb/tech-briefs/manufacturing-and-prototyping/21384-mfs-32715-1 ; accessed 1/20/16. , Feb-2015
Articles in Peer-reviewed Journals	Su CH. " A method of promoting single crystal yield during melt growth of semiconductors by directional solidification." Journal of Crystal Growth. 2015 Jan 15;410:35-8. http://dx.doi.org/10.1016/j.jcrysgro.2014.10.020 , Jan-2015
Articles in Peer-reviewed Journals	Su C-H. "Thermal conductivity, electrical conductivity, and thermoelectric properties of CdTe and Cd(0.8)Zn(0.2)Te crystals between room temperature and 780oC." AIP Advances. 2015 May;5(5):051778. http://dx.doi.org/10.1063/1.4921025 , May-2015